

# Finance and accounting (A Level)

## A Level Business

### Financial statements

**Financial statements** 财务报表 are the official records that show how a business is doing with money. You study two of them: the income statement and the statement of financial position. By law, companies must publish these each year.

### The income statement

The **income statement** 利润表 shows the **revenue** 营业收入 and costs of a business over a period (usually a year), and the profit left at each stage.

$$\text{gross profit} = \text{revenue} - \text{cost of sales}$$

- **cost of sales** 销售成本 is the direct cost of the goods that were sold.
- **gross profit** 毛利润 is what is left after taking cost of sales from revenue.
- after taking off **expenses** 费用 (such as rent, wages and marketing), you get the **operating profit** 营业利润.
- after interest and tax, what is left is the **net profit** 净利润 (the profit for the year).

### The statement of financial position

The **statement of financial position** 财务状况表 (also called the **balance sheet** 资产负债表) is a photo of what the business owns and owes on one day.

- **non-current assets** 非流动资产—items kept for more than a year, such as buildings and machines.
- **current assets** 流动资产—items that become cash within a year, such as inventory and money owed by customers.
- **current liabilities** 流动负债—debts due within a year.
- **non-current liabilities** 非流动负债—debts due after more than a year, such as a long loan.
- **equity** 所有者权益—the money the owners have put in, plus profit kept in the business.

The two sides always balance: what the firm owns is funded by what it owes plus the owners' equity.

### Depreciation

**Depreciation** 折旧 is the way the cost of a non-current asset is spread over the years it is used, instead of all in the year it was bought. For example, a \$10,000 machine used for five years may lose \$2,000 of value each year. Depreciation lowers the asset's value on the statement of financial position and counts as an expense on the income statement, so profit is not overstated.

### How stakeholders use financial statements

Different **stakeholders** 利益相关者 read these statements for different reasons:

- owners and investors check the profit and the return on their money.
- lenders check whether the firm can repay its loans.
- managers use them to make decisions and set budgets.
- the government checks the tax due; suppliers check the firm can pay.

## Ratio analysis

A single number means little on its own. **Ratio analysis** 比率分析 links two figures from the statements to judge performance, and lets you compare one year with another, or one firm with another.

## Profitability ratios

**Profitability** 盈利能力 ratios show how good the firm is at turning sales and capital into profit.

$$\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}} \times 100\%$$

$$\text{operating profit margin} = \frac{\text{operating profit}}{\text{revenue}} \times 100\%$$

A higher **gross profit margin** 毛利率 or **operating profit margin** 营业利润率 means more profit is kept from each sale. The key overall measure is **return on capital employed** 资本回报率 (ROCE), which compares profit with the money invested:

$$\text{ROCE} = \frac{\text{operating profit}}{\text{capital employed}} \times 100\%$$

Here **capital employed** 所用资本 is the total long-term money in the business. A higher ROCE means the money is being used well.

## Liquidity ratios

**Liquidity** 流动性 means how easily a firm can pay its short-term debts. Two ratios test it.

$$\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$$

$$\text{acid test ratio} = \frac{\text{current assets} - \text{inventory}}{\text{current liabilities}}$$

The **current ratio** 流动比率 of about 1.5–2 is usually safe. The **acid test ratio** 速动比率 is stricter because it removes inventory, which can be slow to sell. Too low is risky; too high may mean cash is sitting idle.

## Efficiency and gearing ratios

**Efficiency** 效率 ratios show how well the firm uses its resources —for example, **inventory turnover** 存货周转率 measures how many times a year the firm sells and replaces its stock. Faster turnover usually ties up less cash.

**Gearing** 杠杆比率 shows how much of the firm's long-term money comes from borrowing rather than from owners:

$$\text{gearing} = \frac{\text{non-current liabilities}}{\text{capital employed}} \times 100\%$$

High gearing means heavy borrowing —risky if interest rates rise, but it can boost returns when times are good.

## Limitations of ratio analysis

Ratios are useful but limited. They use past data, which may not predict the future. They ignore non-money factors like staff morale and brand strength. And a fair comparison needs firms of similar size in the same industry.

## Investment appraisal

**Investment appraisal** 投资评估 is the study of whether a big spending project (such as a new machine or factory) is worth it. Three methods are used.

### Payback, ARR and NPV

- the **payback period** 回收期 is the time it takes for the project's cash inflows to repay the initial cost. A shorter payback is safer.
- the **accounting rate of return** 会计回报率 (ARR) shows the average yearly profit as a percentage of the money invested:

$$\text{ARR} = \frac{\text{average annual profit}}{\text{initial investment}} \times 100\%$$

- the **net present value** 净现值 (NPV) recognises that money in the future is worth less than money today. It uses **discounting** 折现 to shrink future cash flows back to today's value, then takes away the initial cost. A positive NPV means the project adds value.

Each method has strengths: payback is simple and focuses on risk; ARR shows profitability; NPV is the most complete but needs a chosen discount rate.

## Finance and accounting strategy

At the top level, managers use financial data and budgets to choose between strategies. Before a big decision, they ask: can we afford it, how will it affect profit, cash flow and gearing, and what is the likely return? Financial information turns a risky guess into a reasoned choice — but numbers are only part of the picture, and must be weighed with the market, the staff and the firm's objectives.